

SEAN CHU

Software Engineer/ Developer

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PROFILE

A highly skilled Software Engineer with over 4 years of experience in developing and optimizing secure, scalable applications using Java, Python, C#, and .NET. Proven track record in enhancing business operations, improving user interfaces, and integrating complex systems across industries such as finance and healthcare. Expertise in full-stack development, cloud deployments on AWS, and driving software efficiency through CI/CD practices. Strong collaborator with a focus on delivering high-performance solutions.

CORE COMPETENCIES

Programming Languages: Java, Python, C#, .NET, C++, JavaScript, TypeScript, SQL

Web Development: HTML/CSS, Flask, Bootstrap, MVC (Model-View-Controller), React

Software Development Tools: Git, GitHub, Docker, Jenkins

Database Management: MySQL, PostgreSQL

Software Engineering Practices: Software Development Lifecycle (SDLC), Continuous Integration/Continuous Deployment (CI/CD)

Project Management: Sprint planning, Feature development/testing, Agile methodologies

Cloud: AWS (EC2, S3, and RDS)

EXPERIENCE

Mastercard, NY

Feb 2023 – Current

Software Engineer

- Enhanced Mastercard Track suite's core features, improving transaction security and operational efficiency suite using Python, .NET, and JavaScript to streamline business operations and facilitate secure B2B transactions globally.
- Implemented RESTful APIs and microservices using Spring Boot and Flask, enabling seamless data integration between internal and external systems.
- Improved supply chain visibility by creating advanced data analytics dashboards using TypeScript, HTML/CSS providing clients with real-time insights into their global supply chain networks.
- Collaborated with cross-functional teams to integrate C# applications with existing security and data management frameworks, improving overall system efficiency and performance.
- Designed and optimized database schemas in MySQL and PostgreSQL, reducing data retrieval latency by up to 30% in high-traffic environments.
- Built and optimized user interfaces using React, enhancing application performance and user experience.
- Established automated CI/CD pipelines using Jenkins, Git, and Docker, significantly reducing deployment times and ensuring consistent software delivery.
- Deployed and managed scalable applications on AWS, utilizing services like EC2, S3, and RDS to enhance performance and ensure high availability.

Epic Systems, Madison, WI

Feb 2020 – Jan 2023

Software Engineer

- Led the development and successful integration of the "EHI" project, which connected the Dorothy and Comfort applications with a major section of the Epic Database, enhancing user access and garnering positive feedback from stakeholders.
- Directed team efforts to comply with new federal healthcare regulations, significantly improving patient access to Electronic Health Records (EHR) via Epic's SQL database system.
- Lead the design, development, and implementation of robust, scalable, and high-performance web applications using .NET framework, ASP.NET MVC.
- Spearheaded the redesign and development of critical applications using C# and .NET, focusing on scalability, security, and performance to meet evolving business needs.
- Enhanced user interfaces using advanced frontend technologies like TypeScript and Bootstrap, improving user experience and application responsiveness. Introduced responsive design principles to ensure optimal viewing across multiple devices.
- Led front-end development efforts to incorporate modern UI/UX practices, utilizing CSS, JavaScript, and React to create intuitive and dynamic user interfaces for complex healthcare applications.

PROJECTS

Discord Bot: Developed an interactive Discord bot that enhances user engagement through automated tasks, moderation tools, and customizable commands for community management.

Falling Leaf Simulator: Created a visually captivating simulation of falling leaves using advanced physics algorithms to mimic natural movements, providing a serene user experience.

EDUCATION

Bachelor of Applied Science in Computer Science (BASc)

2019

Stony Brook University – The State University of New York, New York, NY